e	f	f^2	
f	f^2	e	
f^2	e	f	

} we see closure and inverses ✓

There is also symmetry along the diagonal, so H is Abelian.

Ex. Consider $G \in (\mathbb{Z}, +)$.

What is the smallest subgroup of $\mathbb{Z}$ containing 6? Let's call this the subgroup generated by 6, $\langle 6 \rangle = \{\ldots, -2\cdot 6, -1\cdot 6, 0, 1\cdot 6, 2\cdot 6, \ldots\}$. ↑ inverses need to be here too) (id needs to be there)

$\{6, -6, 18, -18, 24, 30, \ldots\}$

$= 6\mathbb{Z}$ ← the multiples of 6

$|\langle 6 \rangle| = \infty = |G|$

Abstract Algebra 2/1/26

Reminders: professor office hours will be tomorrow Friday, 8 AM - 10 AM, and also next Friday

HW is due Sundays this week and next.

2/2/26

→ A CSUSM prospecting professor will teach our class this Tuesday, as the majority of our class time is spent over topic of their choosing. Professor Smith will send out survey later on to what we thought of their lesson.

Recall: $h \times \cdots \times h = h \cdot \ldots \cdot h = h^n$.

$2^7 = 128$ 7 times

$h^2 \cdot h^3 = \underbrace{h \cdot \ldots \cdot h}_{7 \text{ times}} \cdot \underbrace{h^3 \cdot h^4}_{\text{twice}} = h^{2+3} = h^5$

$\mathbb{Z} \subset \mathbb{Q}$ subgroup under addition

Th^m 3.1 (One Step Subgroup Test) Let G be a group and H a nonempty subset. Then H is a subgroup iff whenever $a, b \in H$, then $(ab^{-1}) \in H$.

Cor. (Finite) If G is a group and $\emptyset \neq H \subseteq G$, then $H \leq G \iff (ab)^{-1} \in H$ whenever $a, b \in H$.

Proof: Consider $(H, \cdot)$ where $\cdot$ is the binary operation from G.

Suppose $ab^{-1} \in H$ whenever $a, b \in H$.

We want to show $H \leq G$. First, since $H \neq \emptyset$, then we have elements in H.

So, we can take $x \in H$. By hypothesis, $\forall a, b \in H$, $ab^{-1} \in H$, so when $a = b = x \in H$, then $x x^{-1} \in H$. Since $x x^{-1} = e$, then $e \in H$. So, the identity is in H. ✓

To show that H is associative, we note that G is associative. Namely, let $a, b, c \in H$. Since $H \subseteq G$, then $a, b, c \in G$. Since G is a group (and thus is associative), then $(ab)c = a(bc) \in G$.

To say all, make sure $(ab)c, a(bc) \in H$, which is to say that H is closed.

For inverses, let $x \in H$. Since $e \in H$ also, then $(ex^{-1}), (x e^{-1}) \in H$, by hypothesis. Since $ex^{-1} = x^{-1}$, then $x^{-1} \in H$. Thus H is closed under taking inverses, since $x \in H$ implied $x^{-1} \in H$.

For closure, let $x, y \in H$. Since $y \in H$, then $y^{-1} \in H$. Since $x, y^{-1} \in H$, $x(y^{-1}) \in H$ by hypothesis. $(y^{-1})^{-1} = y$ because inverses are unique; thus $(x y) \in H$. So, $x, y \in H$ implies $(xy) \in H$, meaning H is closed under its group operation.

Now, since we've shown that H is closed, $abc = (ab)c = a(bc) \in H$. ✓ ∎

Exercise: prove the other direction

If $H \leq G$, then $H \neq \emptyset$ & $ab^{-1} \in H$ whenever $a, b \in H$

Two-Step Subgroup Test

Let G be a group and H a nonempty subset.

H is a subgroup of G iff

1) H is closed under the group opp. of G

2) H is closed under taking inverses.

Proof: ⟸ Suppose H satisfies 1) & 2).

We will apply the 1-Step Test.

1st, $H \neq \emptyset$ ✓ (by hypothesis)

Let $a, b \in H$. Since H is closed under taking inverses, $b^{-1} \in H$. Since H is closed under the group operation, $ab^{-1} \in H$. So, $a, b \in H$ implies $ab^{-1} \in H$, meaning H satisfies the hypothesis of the 1-Step! Thus, by the 1-Step Test, $H \leq G$. ∎

⟹ (Typo in the book of broken dreams)

Def^n (Overloading Notation) 2/3/26

Define $\langle a \rangle = \{a^n : n \in \mathbb{Z}\}$, for a in a group G. (The cyclic subgroup generated by a)

Claim: $\langle a \rangle \leq G$

Proof: We apply the one-step subgroup test. We see that $a^1 = a \in \langle a \rangle$; so $\langle a \rangle \neq \emptyset$. (Let $x, y \in \langle a \rangle$. By def^n of $\langle a \rangle$, $\exists$ $k, m \in \mathbb{Z}$ s.t. $x = a^k$ and $y = a^m$. Notice $y^{-1} = (a^{-1})^{-1} = a^{-m}$ and since $-m \in \mathbb{Z}$, then $a^{-m} \in \langle a \rangle$. Now, we compute $xy^{-1} = (a^k)(a^{-m}) = a^k a^{-m} = a^{k-m} \in \langle a \rangle$, since $(k-m) \in \mathbb{Z}$. The 1-Step Test lets us conclude $\langle a \rangle \leq G$. ∎

Claim: $\langle a \rangle$ the smallest subgroup containing a is $\{a^n : n \in \mathbb{Z}\}$

"smallest" meaning for any subgroup, K, containing a, then $\langle a \rangle \subseteq K$

Proof: Let H be the smallest subgroup containing a. Since $a \in \langle a \rangle$, we see $H \subseteq \langle a \rangle$. Conversely, let $x \in \langle a \rangle$. By def^n of $\langle a \rangle$, $\exists m \in \mathbb{Z}$ s.t. $x = a^m$.

Case 1: If $m = 0$, $x = a^0 = e$. Since $e \in H$, then $x \in H$.

Case 2: If $m > 0$, then $x = a^m = \underbrace{a \cdot \ldots \cdot a}_{m \text{ times}} \in H$ since H is closed under group opp., $x \in H$.

Case 3: If $m < 0$, then $x = a^m = \underbrace{a^{-1} \cdot \ldots \cdot a^{-1}}_{-m \text{ times}} \in H$.

Since $a^{-1} \in H$ and H is closed under the group opp., So, $x \in H$.

In all cases, $x \in \langle a \rangle$ implies $x \in H$, thus $\langle a \rangle \subseteq H$.

Since we've shown $\subseteq$ & $\supseteq$, then $\langle a \rangle = H$. ∎